

KOUSHIK GARA

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Education

BITS Pilani, Hyderabad

Aug 2019 – May 2023

Bachelor of Engineering in Electrical and Electronics Engineering **CGPA: 8.24**

Focused coursework on Data Structures & Algorithms, Object Oriented Programming

Experience

Biz2X

Jul 2023 – Present

Software Engineer

Noida, India

Rules Engine

- Designed and implemented Rules Engine service using **SpringBoot** and **Redis** that reduced loan processing time by **50%** by automating **80+** business rules spanning eligibility, document collection, fraud detection, risk assessment.
- Optimized evaluation performance by introducing multi-layer caching (**Redis + in-memory**) for attribute metadata and rule definitions, cutting Meta Service and database calls by **80%**.
- Scaled to process **150+** evaluations/sec with **<200ms P99** latency using a hybrid (**batch+realtime**) evaluation model, handling **1M+ evaluations**.

Metadata Ecosystem

- Built a centralized metadata management system using **SpringBoot** and **MongoDB** to store and interact with client attributes handling **3000+** metadata entries for each client.
- Applied **Abstract Factory design** for entity resolution and made availability-over-consistency **CAP trade-offs** for real-time processing, exposed metadata via **idempotent REST APIs**, with **90%** uptime.
- Implemented event sourcing with **Spring Kafka**, enabling real-time metadata sync across **10+ services** with **<5s** propagation delay.

Bank Statement Parser

- Developed a configurable parser service using the **Template Method pattern** and **JAXB** for **XML** based parsed data, streamlining the creation of parsing configurations and enabling support for **80+** bank formats.
- Engineered an intelligent fallback system that automatically detects and routed only un-parsable documents (**25%**) to third-party services, reducing annual costs by **\$24K (75% savings)**.
- Enabled automatic updates and seamless addition of new configurations without code changes, reducing deployment overhead and cutting update time by **66%**.

Software Intern

Jul 2022 – Jan 2023

Lending Partner Integration

- Devised routing system using the **Factory pattern** to route requests to specific partner implementations, improving maintainability by **40%**.
- Executed **exponential backoff** and **retry** mechanisms for critical partner API calls using **Spring Retry**, reducing request failures by **65%** and cutting retry-related system load by **30%**.
- Employed the **Chain of Responsibility pattern** to reroute applications to alternate partners in case of failures, significantly increasing system resilience.

Skills

- Languages:** Java, Python, JavaScript(ES6+), TypeScript, HTML, CSS, Bootstrap.
- Databases:** MongoDB, PostgreSQL, Redis
- Frameworks and Skills :** Spring Boot, Microservices, Junit, REST APIs, Spring Kafka, Spring Retry, Git, Agile.
- Dev Tools :** GitHub, Git Lab, IntelliJ, Maven, VS Code, Postman, Jira, Docker, Jenkins, AWS

Awards & Achievements

- Winner of **xTech Hackathon** for building a document parser that accurately identifies and extracts key data from diverse **U.S. government tax statements** with full validation.
- Recipient of the **Functional Award** in the **Innovations Team** at **Biz2X** for driving innovative solutions through continuous learning and impactful team contributions.